

**BELIMO**

# Steam Valve Sizing Tool

*(Belimo Steam Valve Sizing Tool.xlsm)*

## USER MANUAL

Release 3.4

Prepared for Belimo customers, contractors, and internal staff  
Belimo U.S., Inc. | 33 Turner Road, Danbury, CT 06810  
Questions? Contact your Belimo Regional Application Consultant.

## Table of Contents

1. Introduction
  2. Getting Started
  3. The Calculator Sheet — Step by Step
  4. Feedback, Release Information & Language
  5. The 1/3-2/3 Split-Range Valve Calculator
  6. The Summary Sheet
  7. The Tool Box — Extra Engineering Calculators
  8. Steam Pipe Capacity Reference (SteamPipeSize Sheet)
  9. Valve Database Reference (Valves Sheet)
  10. Standalone Pipe & Valve Velocity Checker (Velocity Sheet)
  11. The Engineering Behind the Tool, in Plain Language
  12. Troubleshooting & Getting Help
- Appendix A — All Dropdowns, One Table

# 1. Introduction

The Belimo Steam Valve Sizing Tool (the “tool,” file name Belimo Steam Valve Sizing Tool.xlsm) is an Excel-based calculator that helps you pick the right steam control valve for a heating coil, fin-tube radiator, heat exchanger, or other steam load. You tell it what the application is and what the steam conditions are, and it works out the valve size (Cv or Kv) you need, checks the flow for choked (critical) flow conditions, suggests the closest Belimo valve, and checks pipe and valve velocities so you don't end up with a noisy or oversized installation.

This manual is written for two audiences at once: engineers and contractors sizing a valve for a real job, and Belimo staff who need to explain or troubleshoot the tool for a customer. Wherever the tool uses an engineering term (like “choked flow” or “critical pressure ratio”), this manual explains it in plain language first, then gives the technical detail for anyone who wants it.

## Who this manual covers

Whether you are a contractor filling in numbers for a single job, or a Belimo application engineer walking a customer through the tool over the phone, every dropdown, button, and result on every sheet is documented here — in order, with screenshots.

## 1.1 What the tool does, in one paragraph

You enter the application type, whether the valve is on/off or modulating, the steam flow rate, the inlet and outlet pressures, and your elevation/location (for barometric pressure correction). The tool calculates the required valve coefficient (Cv or Kv), tells you whether the flow is choked (a special condition explained in Section 11), recommends the closest Belimo steam valve from the product line, and — once you tell it your pipe sizes — checks that steam velocity through the valve and piping stays in a safe, quiet range.

## 1.2 What's in this manual

- Section 2 – Getting started: opening the file, enabling macros, and the About screen
- Section 3 – The Calculator sheet: every input, dropdown, and result, step by step.
- Section 4 – Feedback, release information, and changing the tool's language.
- Section 5 – The 1/3-2/3 split-range valve calculator.
- Section 6 – The Summary sheet, for logging multiple sizing results.
- Section 7 – The Tool Box, a set of extra unit-conversion and engineering calculators.
- Section 8 – The Steam Pipe Capacity reference table.
- Section 9 – The Valve database reference.
- Section 10 – The standalone Pipe & Valve Velocity checker.
- Section 11 – The engineering behind the tool, explained simply.
- Section 12 – Troubleshooting and getting help.

- Appendix A – Every dropdown in the tool, in one table, for quick reference.

## 2. Getting Started

### 2.1 Opening the file

Double-click Belimo Steam Valve Sizing Tool.xlsm to open it in Excel. Because the tool uses macros (VBA) to run its calculations, Excel will show a yellow “Security Warning” bar under the ribbon the first time you open it.

- Click “Enable Content” (or “Enable Macros”) on that yellow bar.
- If the file was downloaded from email or the internet, Windows may also ask you to confirm you trust it — this is normal for any macro-enabled workbook and is safe for this file.

#### Why macros are required

All of the sizing math, the valve auto-selection, the unit conversions, and the help tooltips are built with VBA (Excel's built-in programming language). If macros are disabled, the buttons and automatic calculations on the Calculator sheet will not work.

### 2.2 The About screen

Click the small (i) About button near the top of the Calculator sheet, next to the Close Tool button, to open the About screen. It shows the tool's name, its version and release date, an internal-use notice, the tool's disclaimer, and who maintains the tool. It also has three buttons – Check for Updates, Download Fresh Copy and Download User Manual (see Section 4.2). The disclaimer is there for information: the tool's results are meant to guide valve selection and should always be verified against the actual system design conditions.

### 2.3 Where you'll spend most of your time

Almost everything you need for day-to-day sizing lives on one sheet: Calculator. The other sheets (OneThirdTwoThird, Summary, Tools, SteamPipeSize, Valves, Velocity, and the language/update sheets) are all reference tools or specialized calculators that support the main Calculator sheet, and are covered in Sections 5–10. You switch between them using the sheet tabs along the bottom of the Excel window.

### 3. The Calculator Sheet — Step by Step

This is the heart of the tool. Fill in the highlighted input cells from top to bottom, and the results fill themselves in as you go. The screenshot below shows the sheet with no inputs entered yet, so you can see where everything is.

The screenshot displays the 'Belimo Steam Valve Sizing Tool' spreadsheet. The interface includes a title bar with 'App' and a formula bar. The spreadsheet is divided into several sections:

- APPLICATION DATA:** Includes dropdowns for 'Application [APP]' and 'On/Off or Modulating Control? [Ctrl]'. Buttons for 'SHOW / HIDE STEPS' and 'CLOSE TOOL' are present.
- STEAM DATA:** Includes input fields for 'FLOWRATE [Q]', 'Supply INLET Pressure [Pi]', 'OUTLET Pressure [Po]', and 'Atmospheric Pressure [AP]'. Units are specified as LB/HR, PSIG, and Default.
- Cv CALCULATOR RESULTS:** Displays calculated values for 'Converted Flowrate [Q]', 'Downstream Psig [DSP]', 'PSI Drop [h]', 'Inlet Pressure [Pi]', 'Outlet Pressure + AP [Poa]', and 'Flow Coefficient Result [Cv]'. Units include LB/HR, psig (gauge), psig (gauge), psig (gauge), psia (absolute), and Cv.
- CORRECTED PRESSURE DROP [H] & [Q] BASED ON VALVE Cv:** Displays calculated values for 'Selected Valve Input [Cv]', 'Corrected Pressure Drop', 'Corrected Drop Ratio ( $\Delta P_{corr}/P1$ )', 'Flow Regime', 'Inlet Absolute Pressure (P1\_abs)', 'Corrected Outlet Abs Pressure (P2\_)', 'Corrected Flow Output', 'Required [Cv]', 'Difference [Cv]', and 'Margin [%]'. Units include Cv, psi, %, 0, psia, lb/hr, and Cv.

A diagram of a 'G6 Series Globe Valve' is shown in the center, illustrating the valve's components and flow direction.

Figure 1 – The Calculator sheet, blank / default state.

#### 3.1 Application

Cell D6. This tells the tool what kind of steam load you're sizing the valve for. This mainly affects which help text and defaults are shown — the sizing math itself is the same steam-flow calculation regardless of application.

| Field / Dropdown | Where it is               | Choices                                  |
|------------------|---------------------------|------------------------------------------|
| Application      | Calculator sheet, cell D6 | Coil, Fin Tube, Heat Exchanger, Radiator |

#### 3.2 Control (Ctrl)

Cell D8. Tells the tool whether the valve will be a simple open/closed (on/off) valve or a modulating (throttling, variable-position) valve. Modulating applications generally call for equal-percentage valves for good control across the full stroke, while on/off applications are less sensitive to valve characteristic.

| Field / Dropdown | Where it is               | Choices            |
|------------------|---------------------------|--------------------|
| Control          | Calculator sheet, cell D8 | On/Off, Modulating |

#### 3.3 Flow Rate and Flow Unit

Cell D12 (value) with its unit dropdown. Enter the design steam flow rate for the load, then choose the unit it's expressed in. The tool converts everything internally to lb/hr before running the sizing calculation, so you can enter the number in whichever unit you have on hand — a boiler rating in BTU/HR, a coil rating in EDR, or a straightforward lb/hr or kg/h flow.

| Field / Dropdown | Where it is                                | Choices                       |
|------------------|--------------------------------------------|-------------------------------|
| Flow Unit        | Calculator sheet, cell D12 (unit selector) | BTU/HR, EDR, LB/HR, MBH, KG/h |

### 3.4 Inlet Pressure (PI) and Outlet Pressure (PO)

Cells D13 and D14 (with unit selectors at D13/T-column and D14/P-column). Enter the steam pressure just upstream of the valve (inlet, PI) and just downstream of the valve (outlet, PO). Both can be entered in PSIG (pounds per square inch, gauge) or BAR — pick whichever matches your job data.

| Field / Dropdown          | Where it is                | Choices   |
|---------------------------|----------------------------|-----------|
| Inlet Pressure Unit (PI)  | Calculator sheet, cell D13 | PSIG, BAR |
| Outlet Pressure Unit (PO) | Calculator sheet, cell D14 | PSIG, BAR |

The difference between these two pressures (the pressure drop across the valve) is what drives steam through the valve — it's the single biggest factor in how large a valve you need. See Section 11 for why the ratio between PO and PI (not just the difference) also matters.

### 3.5 Location / Barometric Pressure (AP)

Cell D15. Because gauge pressure readings (PSIG) are relative to local atmospheric pressure, and atmospheric pressure changes with elevation, the tool needs to know roughly where the installation is so it can correct PSIG readings to absolute pressure correctly. Pick your U.S. state (or “Manual / Other” to type your own value, or “Default” to use standard sea-level atmospheric pressure) from the dropdown.

| Field / Dropdown                                                                                                                                                                                                                             | Where it is                | Choices                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------|
| Location (Barometric Pressure)                                                                                                                                                                                                               | Calculator sheet, cell D15 | Default, Manual / Other, plus all 50 U.S. states and additional Canadian provinces (74 entries total — see Appendix A) |
| <b>In plain English</b><br>Think of this dropdown as telling the tool “how high up am I,” because a gauge pressure reading of 0 PSIG means something slightly different in Denver than it does in Boston. Picking your state gets this right |                            |                                                                                                                        |

automatically; “Manual / Other” lets you type an exact local barometric pressure if you have one.

### 3.6 Coefficient Type (Cv or Kv)

Cell D28. Choose whether you want the required valve size expressed as Cv (the U.S./Imperial flow coefficient) or Kv (the metric flow coefficient). The tool calculates and displays the result in whichever one you pick.

| Field / Dropdown                                                                                                                                                                                                                                                                                                                                                                                     | Where it is                | Choices |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------|
| Coefficient Type                                                                                                                                                                                                                                                                                                                                                                                     | Calculator sheet, cell D28 | Cv, Kv  |
| <p><b>Known display quirk</b></p> <p>Toggling this dropdown updates the calculated value correctly, but on the 1/3-2/3 split-range calculator (Section 5) the small “Cv/Kv” label next to the result does not visually update when you switch — it's a cosmetic label issue only, not a calculation error. The number itself is correct for whichever type you selected on the Calculator sheet.</p> |                            |         |

### 3.7 Reading the results

As soon as Application, Control, Flow, Inlet Pressure, and Outlet Pressure are all filled in, the tool calculates automatically — there's no “Calculate” button to click for the main sizing result. The screenshot below shows a completed example: a Coil application, Modulating control, 2,500 lb/hr flow, 60 PSIG inlet, 10 PSIG outlet.

AP: 14.7

**BELIMO**

**Belimo Steam Valve Sizing Tool**

SHOW / HIDE STEPS CLOSE TOOL

**APPLICATION DATA**

Application [APP] Coil

On/Off or Modulating Control? [Ctrl] Modulating

**STEAM DATA**

FLOWRATE [Q] 2500.00 LB/HR

Supply INLET Pressure [PI] 60 PSIG

OUTLET Pressure [PO] 10 PSIG

Atmospheric Pressure [AP] 14.7

RESET CALCULATE DISPLAY CV'S ADD TO SUMMARY

**G6 Series Globe Valve**

**Cv CALCULATOR RESULTS**

Converted Flowrate [Q] 2500.00 LB/HR

Downstream Psig [DSP] 10.00 psig (gauge)

PSI Drop [h] 50.00 psig (gauge)

Inlet Pressure [PI] 60.00 psig (gauge)

Outlet Pressure + AP [Poa] 24.70 psia (absolute)

Flow Coefficient Result [Cv] 20.66 Cv

**CRITICAL CALCULATION**

$Cv = Q / (1.62 * P1\_abs)$

$Cv = 2500.0 / (1.62 * 74.700)$

Selected valve Cv = 24.000 (closest = required) - B2100VB-024 (1-in).

**CORRECTED PRESSURE DROP [H] & [Q] BASED ON VALVE Cv**

Selected Valve Input [Cv] 24 Cv

Corrected Pressure Drop 17.69 psi

Corrected Drop Ratio ( $\Delta P\_corr/P1$ ) 23.68 %

Flow Regime Subcritical

Inlet Absolute Pressure (P1\_abs) 74.7 psia

Corrected Outlet Abs Pressure (P2\_ 57.00817066 psia

Corrected Flow Output 2,904.34 lb/hr

Required [Cv] 20.66 Cv

Difference [Cv] 3.34 Cv

Margin [%] 16.17 %

Valve Selection B2100VB-024 (1-in) - Cv 24.0

Figure 2 – A worked example, fully filled in with results calculated.

In this example the tool calculated a required Cv of 20.7 and automatically selected the B2100VB-024 (Cv 24.0) as the closest match – the smallest valve whose Cv is at least what's required, so it is never undersized. Section 3.11 explains the red velocity result.

### 3.7.1 Manually overriding the selected valve

If you want to size a specific valve model yourself instead of using the tool's automatic pick, use the valve selection dropdown (cell J23) to choose from the full Belimo steam valve product line directly.

| Field / Dropdown                 | Where it is                | Choices                                                                                                                                                                                                |
|----------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selected Valve (manual override) | Calculator sheet, cell J23 | -Select-, plus all Belimo steam valve models with size and Cv shown, e.g. "B2050VS-01 (0.5-in) - Cv 1.0" ... "B2150VS-055 (1.5-in) - Cv 55.0" and so on (60 entries — see the Valves sheet, Section 9) |

The tool checks the pressure rating of whatever valve you pick against the job's pressure conditions. If you select a valve that isn't rated for the pressure you've entered, a message box will warn you and ask you to choose a different model — this guards against accidentally specifying an undersized-pressure-class valve.

## 3.8 Suggested Valve & Pipe Size / Pipe & Valve Velocity Check

Scroll down on the Calculator sheet and you'll find two more result panels that appear once the main sizing is complete: a Suggested Valve & Pipe Size summary, and a Pipe & Valve Velocity Check section where you tell the tool your actual pipe sizes so it can verify velocity.

| AP |                                                                                 |                                     |                 |   |   |                                                                                                                                                      |               |       |                              |   |   |   |   |
|----|---------------------------------------------------------------------------------|-------------------------------------|-----------------|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|------------------------------|---|---|---|---|
| A  | B                                                                               | C                                   | D               | E | F | G                                                                                                                                                    | H             | I     | J                            | K | L | M | N |
| 20 | Cv CALCULATOR RESULTS                                                           |                                     |                 |   |   | CORRECTED PRESSURE DROP [H] & [Q] BASED ON VALVE Cv                                                                                                  |               |       |                              |   |   |   |   |
| 21 |                                                                                 |                                     |                 |   |   |                                                                                                                                                      |               |       | Valve Selection              |   |   |   |   |
| 22 |                                                                                 |                                     |                 |   |   |                                                                                                                                                      |               |       | B2100VB-024 (1-in) - Cv 24.0 |   |   |   |   |
| 23 | Converted Flowrate [Q]                                                          | 2500.00                             | LB/HR           |   |   | Selected Valve Input [Cv]                                                                                                                            | 24            | Cv    |                              |   |   |   |   |
| 24 | Downstream Psig [DSP]                                                           | 10.00                               | psig (gauge)    |   |   | Corrected Pressure Drop                                                                                                                              | 17.69         | psi   |                              |   |   |   |   |
| 25 | PSI Drop [h]                                                                    | 50.00                               | psig (gauge)    |   |   | Corrected Drop Ratio ( $\Delta P_{corr}/P1$ )                                                                                                        | 23.68         | %     |                              |   |   |   |   |
| 26 | Inlet Pressure [Pi]                                                             | 60.00                               | psig (gauge)    |   |   | Flow Regime                                                                                                                                          | Subcritical   |       |                              |   |   |   |   |
| 27 | Outlet Pressure + AP [Poa]                                                      | 24.70                               | psia (absolute) |   |   | Inlet Absolute Pressure (P1_abs)                                                                                                                     | 74.7          | psia  |                              |   |   |   |   |
| 28 | Flow Coefficient Result [Cv]                                                    | 20.66                               | Cv              |   |   | Corrected Outlet Abs Pressure (P2_                                                                                                                   | 57.00817066   | psia  |                              |   |   |   |   |
| 29 | Critical Calculation                                                            |                                     |                 |   |   | Corrected Flow Output                                                                                                                                | 2,904.34      | lb/hr |                              |   |   |   |   |
| 30 | $Cv = Q / (1.62 * P1\_abs)$                                                     |                                     |                 |   |   | Required [Cv]                                                                                                                                        | 20.66         | Cv    |                              |   |   |   |   |
| 31 | $Cv = 2500.0 / (1.62 * 74.700)$                                                 |                                     |                 |   |   | Difference [Cv]                                                                                                                                      | 3.34          | Cv    |                              |   |   |   |   |
| 32 | Selected valve Cv = 24.000 (closest = required) - B2100VB-024 (1-in).           |                                     |                 |   |   | Margin [%]                                                                                                                                           | 16.17         | %     |                              |   |   |   |   |
| 33 |                                                                                 |                                     |                 |   |   |                                                                                                                                                      |               |       |                              |   |   |   |   |
| 34 | Flow is choked. P2_abs / P1_abs = 0.331 which is less than or equal to 0.58     |                                     |                 |   |   | Selected valve Cv 24.0 passes design flow without choking (16.2% Cv margin). Adequately sized for capacity.                                          |               |       |                              |   |   |   |   |
| 35 | Critical pressure drop reached. Critical downstream pressure limit = 28.63 psig |                                     |                 |   |   | Actual Pressure Drop = 50. psi<br>Original Outlet Absolute Pressure = 24.7 psia<br>Mach 0.799 - Fail                                                 |               |       |                              |   |   |   |   |
| 36 |                                                                                 |                                     |                 |   |   | $V(ft/s) = (Flow \times SpecificVolume\_outlet) / (3600 \times PortArea/144)$<br>$V = (2904.3 \times 7.531) / (3600 \times 0.680/144) = 1286.5 ft/s$ |               |       |                              |   |   |   |   |
| 37 | SUGGESTED VALVE & PIPE SIZE                                                     |                                     |                 |   |   | PIPE & VALVE VELOCITY CHECK                                                                                                                          |               |       |                              |   |   |   |   |
| 38 | Target Velocity (ft/s)                                                          | 80                                  |                 |   |   |                                                                                                                                                      |               |       |                              |   |   |   |   |
| 39 | Valve Size & Cv                                                                 | G240S-N (1.5-in) - Cv 28.0          |                 |   |   | Valve Type                                                                                                                                           | -Select-      |       |                              |   |   |   |   |
| 40 | Valve Travel %                                                                  | 92% (Equal %, R=50)                 |                 |   |   | Steam Type                                                                                                                                           | Dry Saturated |       |                              |   |   |   |   |
| 41 | Calculated Valve Noise (dBA)                                                    | OK                                  |                 |   |   | Supply Pipe Size                                                                                                                                     | -Select-      | inch  |                              |   |   |   |   |
| 42 | Valve Velocity                                                                  | OK                                  |                 |   |   | Outlet Pipe Size                                                                                                                                     | -Select-      | inch  |                              |   |   |   |   |
| 43 | Inlet Pipe Size                                                                 | 4-in Sch. 40 (min. for <=80 ft/s)   |                 |   |   | Valve Size                                                                                                                                           | -Select-      | inch  |                              |   |   |   |   |
| 44 | Outlet Pipe Size                                                                | 5-in Sch. 40 (1 size up from inlet) |                 |   |   | Calculate Velocity                                                                                                                                   |               |       |                              |   |   |   |   |

Figure 3 – Suggested Valve & Pipe Size, and the Pipe & Valve Velocity Check panel.

### 3.8.1 Target Velocity

A small dropdown lets you choose the target design velocity (in ft/s) the tool should check pipe sizing against.

| Field / Dropdown       | Where it is                | Choices     |
|------------------------|----------------------------|-------------|
| Target Velocity (ft/s) | Calculator sheet, cell C38 | 50, 80, 120 |

### 3.8.2 Velocity Check dropdowns

Once you've selected an actual Valve Type and entered your Supply Pipe Size, Outlet Pipe Size, and Valve Size, click Calculate Velocity to check that steam speed through each is within a safe, quiet range.

| Field / Dropdown                      | Where it is                     | Choices                                                                   |
|---------------------------------------|---------------------------------|---------------------------------------------------------------------------|
| Valve Type                            | Calculator sheet, cell H39      | Butterfly, Globe, HTCCV, V-Ball, VS, VSS                                  |
| Steam Type                            | Calculator sheet, cell H40      | Dry Saturated, Wet Saturated, Superheated                                 |
| Supply / Outlet Pipe Size, Valve Size | Calculator sheet, cells H41–H43 | -Select-, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12 (inches) |

If the calculated velocity is too high for the pipe or valve size you've chosen, the tool flags it with a caution message and (as shown in the earlier calculator example) tells you the recommended minimum pipe size instead, so you can correct the selection before it goes on a drawing.

### 3.9 Simple / Detailed view toggle

| Field / Dropdown         | Where it is                | Choices          |
|--------------------------|----------------------------|------------------|
| View (Simple / Detailed) | Calculator sheet, cell C47 | Simple, Detailed |

“Simple” hides the intermediate calculation steps and shows only the inputs and final results — good for a quick job or a customer-facing screen share. “Detailed” shows the full calculation trail (corrected pressure drop, choked-flow check, noise estimate, etc.) so an engineer can verify exactly how the tool arrived at its answer.

### 3.10 On-screen help: the “?” icons and guided walkthrough

Small “?” icons appear next to several inputs (Flow Rate, Inlet Pressure, Outlet Pressure, Barometric Pressure, and the velocity-check fields) and, since release 3.3, next to the key results (Section 3.11). Clicking one shows a short plain-language explanation of that item, right on the sheet, so help is always one click away without leaving the Calculator.

There is also a guided, numbered walkthrough (Steps 1 through 7) that can be shown or hidden as a set of on-sheet callouts pointing at each input in order, which is useful the first time you (or a customer) work through the tool, or when training a new team member.

[Reporting a problem](#)

If the tool ever shows an unexpected error, note what you were doing and what the message said, then use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1) to report it. The more specific you are, the faster it can be tracked down.

### 3.11 Reading the results: the “?” icons on the results side

Release 3.3 added “?” icons to the results side of the Calculator. Click one and a plain-language message opens in the language you have selected. They are there because the two result panels answer different questions, and a large margin, or a red “Fail” beside a valve you did not choose, can look like an error when it is not.

| Icon next to...                       | What it tells you                                                                                                                                                                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Flow Coefficient Result [Cv]</b>   | The Cv (or Kv) needed to pass your flow at the pressures you entered. It is an opening size – not a valve position and not a catalog valve. When the flow is choked it is calculated at the critical pressure drop.                                                                                        |
| <b>Above the Valve Selection box</b>  | The right-hand panel answers the reverse question: what would the valve chosen at J23 do, fully open, at the same flow and inlet pressure? Its pressure drop, flow and outlet pressure are not your operating conditions; a valve larger than needed simply throttles in service.                          |
| <b>Flow Regime</b>                    | Subcritical, or choked (outlet at or below about 58% of the inlet, both absolute). Choked flow depends on the inlet pressure only, so lowering the outlet pressure adds no flow; expect more noise and erosion. The pressure a coil actually sees is set by the coil and its load, not by the valve alone. |
| <b>Required / Difference / Margin</b> | Difference is the selected valve’s Cv minus the required Cv; Margin is that as a percentage of the required Cv. A large margin is not a fault – the next size down may fail the velocity check. A negative margin means the valve is too small.                                                            |
| <b>Target Velocity</b>                | Used only to size the inlet pipe (the outlet pipe is one standard size up). It does not change the required Cv or the valve’s own port-velocity limit.                                                                                                                                                     |
| <b>Valve Size &amp; Cv</b>            | The catalog valve closest to the required Cv among those whose port velocity passes the limit for the Steam Type (wet saturated 40 m/s, dry saturated 150 m/s, superheated 0.5 Mach). Its pressure rating is checked against your inlet pressure.                                                          |
| <b>Valve Travel %</b>                 | An estimate of how far open the valve would need to be at design flow, from a generic equal-percentage curve (the basis, for example “Equal %, R=50”, is shown beside the number). It is not measured data for that valve, so use it as a guide, not as a commissioning setpoint.                          |

#### What the worked example shows

In Figure 2 the job needs a Cv of 20.7. The closest catalog valve (B2100VB-024, Cv 24.0) has a 16% margin but fails the outlet-velocity check (Mach 0.80, in red under the right-hand panel), so the Suggested Valve panel further down picks the G240S-N (1.5-in, Cv 28.0), which passes (a note under that panel says why). Its larger Cv is the price of keeping velocity and noise in range. At design flow it would sit at about 92% of travel (an estimate), so control at that flow is not the concern; minimum load is, and the 1/3-2/3 sheet (Section 5) has the turndown check.

## 4. Feedback, Release Information & Language

Scroll to the bottom of the Calculator sheet and you'll find three more useful sections: a feedback link, the tool's version/update information, and the language selector.

The screenshot shows the bottom portion of a spreadsheet interface. At the top, a red warning message states: "Critical pressure drop reached. Critical downstream pressure limit = 28.63 psig". Below this, a section titled "SUGGESTED VALVE & PIPE SIZE" contains input fields for Target Velocity (ft/s) set to 80, Valve Size & Cv (G240S-N (1.5-in) - Cv 28.0), Valve Travel % (98% (Equal %, R=50)), Calculated Valve Noise (dBA) (OK), Valve Velocity (OK), Inlet Pipe Size (4-in Sch. 40 (min. for <=80 ft/s)), and Outlet Pipe Size (5-in Sch. 40 (1 size up from inlet)). To the right, a "PIPE & VALVE VELOCITY CHECK" section includes dropdowns for Valve Type (-Select-), Steam Type (Dry Saturated), Supply Pipe Size (-Select- inch), Outlet Pipe Size (-Select- inch), and Valve Size (-Select- inch), along with a "Calculate Velocity" button. Below these is a "Result:" field. The "FEEDBACK" section features a "Report a Bug or Give Feedback" button. The "RELEASE INFORMATION" section displays: Release Version 3.1, Release Date 9/23/2026, Update Mode 1, and Version Check Status Success. The "Language" section shows flags for US, UK, DE, and FR, with a "Remember my language" checkbox.

Figure 4 – Feedback, Release Information, and Language sections at the bottom of the Calculator sheet.

### 4.1 Feedback

Click the Report a Bug or Give Feedback button in the Feedback section to open Belimo's bug and feedback form in your web browser. Use it for anything from a typo to a feature request – it's the fastest way to get a change considered for the next release. If you are reporting a problem, tell us what you were doing when it happened (the values you had entered and any message the tool showed); the more specific you are, the faster it can be tracked down.

### 4.2 Release Information

This box shows the current Release Version and Release Date of your copy of the tool, along with the Update Mode and Version Check Status. The tool checks a Belimo-hosted update source each time it opens; "Success" (shown in green) means it confirmed you're on a supported version. If a newer version is available, you'll be offered the chance to view what's new and, if you choose to install it, pick where to save the new copy yourself through a standard Windows Save As dialog – just like saving any other file. Your current copy is never touched or deleted, so there's nothing to lose by trying it.

You don't have to wait for the automatic check. The (i) button near the top of the Calculator sheet, next to the Close Tool button, opens the About screen (Section 2.2). Check for Updates runs the same check on demand. Download Fresh Copy re-downloads the current published version right

away – useful if you suspect your copy is corrupted or missing something, even if you're already on the latest release. Download User Manual saves the current published copy of this manual to a folder you choose; if no manual has been published yet, the tool tells you so.

### 4.3 Language

Click one of the four flag icons to switch the entire tool's labels, buttons, and messages between English, Spanish, Portuguese, and French. Check “Remember my language” to have the tool automatically reopen in your chosen language every time, instead of defaulting back to English. Remembering a language saves the workbook, so if you have inputs on screen the tool asks first: Yes saves the preference and clears the calculator, and No keeps your work and changes the language for this session only.

### 4.4 Closing the tool

The Close Tool button closes this workbook without saving. If no other workbook is open it also closes Excel; if you have other files open, they are left alone.

## 5. The 1/3-2/3 Split-Range Valve Calculator

Some jobs need two valves working together instead of one — a small valve that handles the low end of the load precisely, and a larger valve that opens for peak demand. This is called split-range (or 1/3-2/3) valve staging. The OneThirdTwoThird sheet takes the Cv the Calculator sheet just worked out and splits it into a recommended small/large valve pair.

| A  | B                                          | C                                                                                          | D           | E | F | G |
|----|--------------------------------------------|--------------------------------------------------------------------------------------------|-------------|---|---|---|
| 1  |                                            |                                                                                            |             |   |   |   |
| 2  | 1/3 - 2/3 VALVE CALCULATOR                 |                                                                                            |             |   |   |   |
| 3  |                                            |                                                                                            |             |   |   |   |
| 4  |                                            | 1/3rd                                                                                      | 2/3rd       |   |   |   |
| 5  | Cv 1/3rd - 2/3rd:                          | 6.886255585                                                                                | 13.77251117 |   |   |   |
| 6  |                                            |                                                                                            |             |   |   |   |
| 7  | LB/HR 1/3rd - 2/3rd:                       | 833.3333333                                                                                | 1666.666667 |   |   |   |
| 8  |                                            |                                                                                            |             |   |   |   |
| 9  |                                            |                                                                                            |             |   |   |   |
| 10 | TURNDOWN ANALYSIS (EFFECTIVE RANGEABILITY) |                                                                                            |             |   |   |   |
| 11 |                                            |                                                                                            |             |   |   |   |
| 12 | Suggested Valve (from Calculator)          | G240S-N (1.5-in) - Cv 28.0                                                                 |             |   |   |   |
| 13 |                                            |                                                                                            |             |   |   |   |
| 14 | Minimum / Turndown Flow (lb/hr):           |                                                                                            |             |   |   |   |
| 15 |                                            |                                                                                            |             |   |   |   |
| 16 |                                            |                                                                                            |             |   |   |   |
| 17 |                                            |                                                                                            |             |   |   |   |
| 18 |                                            |                                                                                            |             |   |   |   |
| 19 |                                            |                                                                                            |             |   |   |   |
| 20 |                                            |                                                                                            |             |   |   |   |
| 21 |                                            |                                                                                            |             |   |   |   |
| 22 |                                            |                                                                                            |             |   |   |   |
| 23 |                                            |                                                                                            |             |   |   |   |
| 24 | Recommendation:                            | Enter the Minimum/Turndown Flow above to evaluate the split-range (1/3 - 2/3) requirement. |             |   |   |   |
| 25 | Suggested Split Valves:                    | -Select-                                                                                   |             |   |   |   |
| 26 | Small Valve (Low-Load, 1/3):               | G220S-K (0.75-in) - Cv 7.5 (target 6.89)                                                   |             |   |   |   |
| 27 | Large Valve (Peak, 2/3):                   | G225S-L (1-in) - Cv 14.0 (target 13.77)                                                    |             |   |   |   |
| 28 |                                            |                                                                                            |             |   |   |   |
| 29 |                                            |                                                                                            |             |   |   |   |
| 30 |                                            |                                                                                            |             |   |   |   |
| 31 |                                            |                                                                                            |             |   |   |   |
| 32 |                                            |                                                                                            |             |   |   |   |
| 33 |                                            |                                                                                            |             |   |   |   |
| 34 |                                            |                                                                                            |             |   |   |   |

Figure 5 – The 1/3-2/3 Valve Calculator sheet, with a worked split-range example.

### 5.1 How to use it

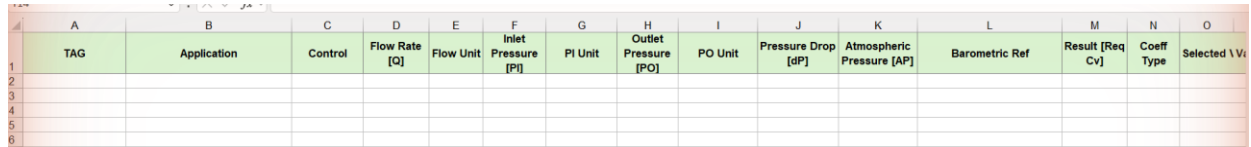
- Size your load normally on the Calculator sheet first — this sheet automatically pulls in the resulting Suggested Valve and its Cv.
- Enter your building's actual Minimum / Turndown Flow (the lowest flow the system realistically needs to deliver, in lb/hr) in the highlighted cell.
- The Turndown Analysis section calculates the effective rangeability needed and tells you, under Recommendation, whether a single valve can handle that turndown or whether a split-range pair is advisable.
- If a split-range pair is recommended, choose Ball or Globe from the Suggested Split Valves dropdown to see the recommended small (low-load, 1/3) and large (peak, 2/3) valve models.

| Field / Dropdown                                                                                                                                        | Where it is                      | Choices               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------|
| Suggested Split Valves                                                                                                                                  | OneThirdTwoThird sheet, cell C25 | -Select-, Ball, Globe |
| <b>Why split the valve into two</b><br>A single valve sized for peak flow often can't control accurately at low flow — it barely has to crack open, and |                                  |                       |

small position changes swing the flow a lot. Splitting the job between a small valve (handling roughly the bottom third of the range) and a large valve (handling the top two-thirds) gives you good control across the whole range instead of just at the top end.

## 6. The Summary Sheet

If you're sizing valves for more than one tag, room, or piece of equipment in the same project, the Summary sheet lets you log each completed calculation as its own row, so you end up with a running record for the whole job instead of re-entering data or losing earlier results.



|   | A   | B           | C       | D             | E         | F                   | G       | H                    | I       | J                  | K                         | L              | M               | N          | O              |
|---|-----|-------------|---------|---------------|-----------|---------------------|---------|----------------------|---------|--------------------|---------------------------|----------------|-----------------|------------|----------------|
| 1 | TAG | Application | Control | Flow Rate [Q] | Flow Unit | Inlet Pressure [PI] | PI Unit | Outlet Pressure [PO] | PO Unit | Pressure Drop [dP] | Atmospheric Pressure [AP] | Barometric Ref | Result [Req Cv] | Coeff Type | Selected Valve |
| 2 |     |             |         |               |           |                     |         |                      |         |                    |                           |                |                 |            |                |
| 3 |     |             |         |               |           |                     |         |                      |         |                    |                           |                |                 |            |                |
| 4 |     |             |         |               |           |                     |         |                      |         |                    |                           |                |                 |            |                |
| 5 |     |             |         |               |           |                     |         |                      |         |                    |                           |                |                 |            |                |
| 6 |     |             |         |               |           |                     |         |                      |         |                    |                           |                |                 |            |                |

Figure 6 – Summary sheet column headers — every field captured per logged calculation.

Once you're happy with a result on the Calculator sheet, use the “Add to Summary” action to copy that job's inputs and results (Tag, Application, Control, Flow Rate, Inlet/Outlet Pressure, Pressure Drop, Atmospheric Pressure, the required Cv, Coefficient Type, and the Selected Valve) into a new row here. Over the course of a project this becomes your schedule of valves — useful for a submittal package or for handing off to purchasing.

## 7. The Tool Box — Extra Engineering Calculators

The Tools sheet is a set of small, independent calculators for everyday HVAC and hydronic math that comes up alongside steam valve sizing — handy even on jobs that aren't primarily about steam.

The Tool Box

Legend:   
 = Input Fields   
 = Output Fields

**Convert Amps to Ohms**

|         |    |
|---------|----|
| Voltage | 24 |
| Amps    | 1  |
| Ohms    | 48 |

**Convert Watts to Volts**

|         |   |
|---------|---|
| Watts   | 5 |
| Amps    | 3 |
| Voltage | 2 |

**Convert Watts to Amps**

|       |      |
|-------|------|
| Watts | 5    |
| Amps  | 0.21 |

**Convert Amps to Watts**

|         |    |
|---------|----|
| Voltage | 24 |
| Amps    | 2  |
| Watts   | 48 |

**Pump Horse Power**

|            |       |
|------------|-------|
| GPM        | 10    |
| Total Head | 30    |
| HP         | 0.076 |

**Cv = GPM / Sq Root of the Delta P**

|             |       |     |
|-------------|-------|-----|
| FLOWRATE    | 35.00 | GPM |
| PSI DROP    | 4.00  | PSI |
| Cv REQUIRED | 17.50 |     |
| Kv          | 10    | 10  |
| = Cv        | 11.6  |     |

**GPM = K x Sq Root of the DeltaP**

|         |       |
|---------|-------|
| K or Cv | 10    |
| DeltaP  | 3     |
| GPM     | 17.32 |

**Coil DeltaP Method**

|                  |         |     |
|------------------|---------|-----|
| Design Flow      | 1000.0  | GPM |
| Measured Coil ΔP | 30.0    | PSI |
| Design Coil ΔP   | 30.0    | PSI |
| Actual GPM       | 1000.00 |     |

**ΔP Calculator = GPM / Cv Squared**

|                 |        |
|-----------------|--------|
| GPM Req'd       | 100.00 |
| Cv of the Valve | 240.00 |
| GPM/Cv =        | 0.42   |
| Adj ΔP =        | 0.17   |

**Valve Authority**

|                     |                   |
|---------------------|-------------------|
| Valve ΔP            |                   |
| Branch ΔP / Coil    |                   |
| Valve Authority     |                   |
| Adj ΔP =            |                   |
| Branch ΔP           |                   |
| Adj Valve Authority | 0.50 & Greater is |

**Valve Flow Rated based on DeltaP**

| Valve Cv | 200    |        |        |        |        |        |        |        |        |        |        |        |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| ΔP       | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     |
| GPM      | 200.00 | 282.84 | 346.41 | 400.00 | 447.21 | 489.90 | 529.15 | 565.69 | 600.00 | 632.46 | 663.32 | 692.82 |
| ΔP       | 8.00   | 565.69 |        |        |        |        |        |        |        |        |        |        |

**Fluid Velocity in FPS based on Flow Rate**

| Pipe Size | 1 | 81.70 | 115.54 | 141.51 | 163.40 | 182.69 | 200.12 | 216.16 | 231.08 | 245.10 | 258.36 | 270.97 | 283.02 |
|-----------|---|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|-----------|---|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

**CvC Calculator**

|            |  |
|------------|--|
| Cv         |  |
| Valve Size |  |
| Pipe Size  |  |
| FPM        |  |
| Cvc=       |  |

**By-Pass Valve**

|              |  |
|--------------|--|
| Total GPM    |  |
| Feet of Head |  |
| In PSI       |  |

Figure 7 – The Tool Box sheet — electrical, flow/Cv, and pipe-sizing helper calculators.

### 7.1 What's included

- Electrical conversions: Amps ↔ Ohms ↔ Watts ↔ Volts, and Pump Horsepower from GPM and Total Head.
- $Cv = GPM / \sqrt{\Delta P}$  and  $GPM = K \times \sqrt{\Delta P}$  — the basic hydronic (water-side) valve sizing formulas, for quick liquid-flow Cv checks outside of the steam calculator.
- Coil ΔP method and ΔP Calculator (GPM/Cv squared) — for estimating a coil's actual operating pressure drop from a design condition.
- Valve Flow Rated based on ΔP, and Fluid Velocity in FPS based on Flow Rate — quick reference tables across a range of ΔP and flow values.
- Valve Authority and Branch/Coil pressure-drop ratio calculators — for checking that a control valve has enough authority over its circuit to control well.
- A CvC (Cv Calculator) and By-Pass Valve calculator for auxiliary sizing questions.

These tools are self-contained on this sheet — they don't feed back into the steam Calculator sheet — so you can use them for a quick side calculation without disturbing your steam sizing job.

## 8. Steam Pipe Capacity Reference (SteamPipeSize Sheet)

This sheet is a reference table, not a calculator you fill in — it's the lookup table the tool itself uses (and that you can use directly) to check how much steam a given pipe size can carry at a given pressure and steam speed.

|    | A                | B                    | C                | D         | E      | F         | G        | H      | I        | J      | K      | L      | M      | N      | O       | P       | Q |
|----|------------------|----------------------|------------------|-----------|--------|-----------|----------|--------|----------|--------|--------|--------|--------|--------|---------|---------|---|
| 1  | Capacity (lb/hr) |                      | Pipe Size (inch) |           |        |           |          |        |          |        |        |        |        |        |         |         |   |
| 2  | Pressure (psi)   | Steam Speed (ft/sec) | 0.5-inch         | 0.75-inch | 1-inch | 1.25-inch | 1.5-inch | 2-inch | 2.5-inch | 3-inch | 4-inch | 5-inch | 6-inch | 8-inch | 10-inch | 12-inch |   |
| 3  | 5                | 50                   | 12               | 26        | 45     | 70        | 100      | 190    | 280      | 410    | 760    | 1250   | 1770   | 3100   | 5000    | 7100    |   |
| 4  | 5                | 80                   | 19               | 45        | 75     | 115       | 170      | 300    | 490      | 710    | 1250   | 1800   | 2700   | 5200   | 7600    | 11000   |   |
| 5  | 5                | 120                  | 29               | 60        | 110    | 175       | 245      | 460    | 700      | 1000   | 1800   | 2900   | 4000   | 7500   | 12000   | 16500   |   |
| 6  | 10               | 50                   | 15               | 35        | 55     | 88        | 130      | 240    | 365      | 550    | 950    | 1500   | 2200   | 3770   | 6160    | 8500    |   |
| 7  | 10               | 80                   | 24               | 52        | 95     | 150       | 210      | 380    | 600      | 900    | 1500   | 2400   | 3300   | 5900   | 9700    | 13000   |   |
| 8  | 10               | 120                  | 35               | 72        | 135    | 210       | 330      | 590    | 850      | 1250   | 2200   | 3400   | 4800   | 9000   | 14400   | 20500   |   |
| 9  | 20               | 50                   | 21               | 47        | 82     | 123       | 185      | 210    | 520      | 740    | 1340   | 1980   | 2900   | 5300   | 8000    | 11500   |   |
| 10 | 20               | 80                   | 32               | 70        | 120    | 190       | 260      | 520    | 810      | 1100   | 1900   | 3100   | 4500   | 8400   | 13200   | 18300   |   |
| 11 | 20               | 120                  | 50               | 105       | 190    | 300       | 440      | 840    | 1250     | 1720   | 3100   | 4850   | 6750   | 1300   | 19800   | 28000   |   |
| 12 | 30               | 50                   | 26               | 56        | 100    | 160       | 230      | 420    | 650      | 950    | 1650   | 2600   | 3650   | 6500   | 10500   | 14500   |   |
| 13 | 30               | 80                   | 42               | 94        | 155    | 250       | 360      | 655    | 950      | 1460   | 2700   | 3900   | 5600   | 10700  | 16500   | 23500   |   |
| 14 | 30               | 120                  | 62               | 130       | 240    | 370       | 570      | 990    | 1550     | 2100   | 3950   | 6100   | 8700   | 16000  | 25000   | 35000   |   |
| 15 | 40               | 50                   | 32               | 75        | 120    | 190       | 260      | 505    | 790      | 1100   | 1900   | 3100   | 4200   | 8200   | 12800   | 18000   |   |
| 16 | 40               | 80                   | 51               | 110       | 195    | 300       | 445      | 840    | 1250     | 1800   | 3120   | 4600   | 6800   | 13400  | 20300   | 28300   |   |

Figure 8 – Steam Pipe Capacity table (lb/hr) by pressure, steam speed, and pipe size.

Find your design steam pressure and target steam speed (ft/sec) in the left-hand columns, then read across to the pipe size column that just meets or exceeds your required flow (in lb/hr) — that's your minimum pipe size. This is exactly the table the Calculator sheet's automatic “min. pipe size” recommendations (Section 3.8) are pulled from, so if you ever want to double check or explain a recommendation to a customer, this is where the numbers come from.

## 9. Valve Database Reference (Valves Sheet)

This sheet holds the complete Belimo steam valve product data the tool draws on for automatic valve selection and for the manual valve-selection dropdown (Section 3.7.1).

|    | A           | B     | C    | D       | E                      | F  | G         | H        | I                 | J                      | K          | L            |
|----|-------------|-------|------|---------|------------------------|----|-----------|----------|-------------------|------------------------|------------|--------------|
|    | Part #      | Model | Size | Cv (US) | Kv (m <sup>3</sup> /h) | PI | ValveType | Ball Dia | Ball Area (sq-in) | Seat/Port Area (sq-in) | Curve Type | Rangeability |
| 1  | B2050VS-01  | VS    | 0.5  | 1.00    | 0.87                   | 35 | VS        | 0.1870   | 0.0275            |                        | Equal %    | 100          |
| 2  | B2050VS-02  | VS    | 0.5  | 2.00    | 1.73                   | 35 | VS        | 0.2810   | 0.0620            |                        | Equal %    | 100          |
| 3  | B2050VS-04  | VS    | 0.5  | 4.00    | 3.46                   | 35 | VS        | 0.3750   | 0.1104            |                        | Equal %    | 100          |
| 4  | B2050VS-15  | VS    | 0.5  | 15.00   | 12.98                  | 35 | VS        | 0.5000   | 0.1963            |                        | Equal %    | 100          |
| 5  | B219VS      | VS    | 0.75 | 30.00   | 25.95                  | 35 | VS        | 0.6800   | 0.3632            |                        | Equal %    | 100          |
| 6  | B220VS      | VS    | 0.75 | 51.00   | 44.12                  | 35 | VS        | 0.8600   | 0.5809            |                        | Equal %    | 100          |
| 7  | B224VS      | VS    | 1    | 43.00   | 37.20                  | 35 | VS        | 0.8700   | 0.5945            |                        | Equal %    | 100          |
| 8  | B225VS      | VS    | 1    | 68.00   | 58.82                  | 35 | VS        | 1.0000   | 0.7854            |                        | Equal %    | 100          |
| 9  | B232VS      | VS    | 1.25 | 48.00   | 41.52                  | 35 | VS        | 1.0000   | 0.7854            |                        | Equal %    | 100          |
| 10 | B239VS      | VS    | 1.5  | 84.00   | 72.66                  | 35 | VS        | 1.2500   | 1.2272            |                        | Equal %    | 100          |
| 11 | B240VS      | VS    | 1.5  | 177.00  | 153.11                 | 35 | VS        | 1.5000   | 1.7671            |                        | Equal %    | 100          |
| 12 | B249VS      | VS    | 2    | 108.00  | 93.42                  | 35 | VS        | 1.5000   | 1.7671            |                        | Equal %    | 100          |
| 13 | B2050VSS-15 | VSS   | 0.5  | 15.00   | 12.98                  | 50 | VSS       | 0.5000   | 0.1963            |                        | Equal %    | 100          |
| 14 | B219VSS     | VSS   | 0.75 | 30.00   | 25.95                  | 50 | VSS       | 0.6800   | 0.3632            |                        | Equal %    | 100          |
| 15 | B224VSS     | VSS   | 1    | 43.00   | 37.20                  | 50 | VSS       | 0.8700   | 0.5945            |                        | Equal %    | 100          |
| 16 | B249VSS     | VSS   | 2    | 108.00  | 93.42                  | 50 | VSS       | 1.5000   | 1.7671            |                        | Equal %    | 100          |

Figure 9 – Valve database: part number, model, size, Cv/Kv, pressure rating, ball/port geometry, valve curve, and rangeability for every valve.

### 9.1 Columns explained

- Part # / Model — the Belimo catalog part number and model family (e.g., VS, VSS, HTCCV).
- Size — valve size in inches.
- Cv (US) / Kv (m<sup>3</sup>/h) — the valve's rated flow coefficient in both unit systems.
- PI — the valve's maximum rated inlet pressure (used for the pressure-rating check described in Section 3.7.1).
- ValveType, Ball Dia., Seat/Port Area — the physical geometry behind the flow coefficient.
- Curve Type — the valve's inherent flow characteristic (Equal % for most of these models, meaning flow changes proportionally with position — well suited to modulating control).
- Rangeability — the ratio between the valve's maximum and minimum controllable flow; this is what feeds the 1/3-2/3 turndown analysis in Section 5.

You generally won't need to edit this sheet — it updates automatically as new valve models are added to a future release — but it's the place to look if you want to see the raw catalog data behind a recommendation.

## 10. Standalone Pipe & Valve Velocity Checker (Velocity Sheet)

This sheet is the same velocity-check logic used inside the Calculator sheet (Section 3.8.2), but available as its own standalone tool — useful if you want to check a pipe/valve velocity for a job that was sized outside this tool, or re-check an as-built installation.

| A  | B                                                                                                                                                                           | C                   | D                            | E                                         | F                         | G               | H                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------|-------------------------------------------|---------------------------|-----------------|-----------------------|
| 1  | <b>Steam Valve Outlet Velocity Check</b>                                                                                                                                    | Valve Type [Select] | Globe                        |                                           | Calculate Outlet Velocity |                 |                       |
| 2  | Steam Quality Override (blank = auto-classify from entropy)                                                                                                                 | Dry Saturated       |                              |                                           |                           |                 |                       |
| 3  | Min Pipe Size (SteamPipeSize table) @ 50 / 80 / 120 ft/s                                                                                                                    | 4-in                | 3-in                         | 2.5-in                                    | Apply Pipe-Size Floor     |                 |                       |
| 4  | Steam Flow                                                                                                                                                                  | P1 Inlet Abs        | P2 Corrected Abs             | Specific Volume Out (ft <sup>3</sup> /lb) | Steam Classification      | Quality (x)     | Speed of Sound (ft/s) |
| 5  | 2493.4 lb/hr                                                                                                                                                                | 79.70 psia          | 75.77 psia                   | 5.761                                     | Dry Saturated             | 1               | 1621.9                |
| 6  |                                                                                                                                                                             |                     |                              |                                           |                           |                 |                       |
| 7  |                                                                                                                                                                             |                     |                              |                                           |                           |                 |                       |
| 8  |                                                                                                                                                                             |                     |                              |                                           |                           |                 |                       |
| 9  |                                                                                                                                                                             |                     |                              |                                           |                           |                 |                       |
| 10 | <b>Selected Valve Result</b>                                                                                                                                                | Cv                  | Port Area (in <sup>2</sup> ) | Area Source                               | Velocity (ft/s)           | Velocity (m/s)  | Mach No.              |
| 11 | Valve (Part#, Size, Type, Cv)<br>B2100VB-024 (1-in, V-Ball, Cv=24.0)                                                                                                        | 24                  | 0.68                         | Measured (Ball Area)                      | 845                       | 257.5           | 0.521                 |
| 12 | RECOMMENDED: B2150VB-055 (1.5-in, Cv=55.0) 91.7 m/s, Mach 0.185 -> Pass                                                                                                     |                     |                              |                                           |                           |                 |                       |
| 13 | Inlet pipe (2-in Sch 40): 163.3 ft/s (49.8 m/s) - CAUTION (above 40 m/s practical limit, up to 76 m/s may be acceptable for short, insulated, dry runs per some standards)  |                     |                              |                                           |                           |                 |                       |
| 14 | Outlet pipe (2-in Sch 40): 171.2 ft/s (52.2 m/s) - CAUTION (above 40 m/s practical limit, up to 76 m/s may be acceptable for short, insulated, dry runs per some standards) |                     |                              |                                           |                           |                 |                       |
| 15 | Recommendation:                                                                                                                                                             |                     |                              |                                           |                           |                 |                       |
| 16 | Catalog Scan (filtered by selected Valve Type)                                                                                                                              |                     |                              |                                           |                           |                 |                       |
| 17 | Part#                                                                                                                                                                       | Size (in)           | Cv                           | Port Area (in <sup>2</sup> )              | Area Source               | Velocity (ft/s) | Velocity (m/s)        |
| 18 | B2100VB-024                                                                                                                                                                 | 1                   | 24                           | 0.68                                      | Measured (Ball Area)      | 845             | 257.5                 |
| 19 | B2150VB-055                                                                                                                                                                 | 1.5                 | 55                           | 1.91                                      | Measured (Ball Area)      | 300.8           | 91.7                  |
| 20 | B2200VB-077                                                                                                                                                                 | 2                   | 77                           | 3                                         | Measured (Ball Area)      | 191.5           | 58.4                  |
| 21 | B6300VB-207                                                                                                                                                                 | 3                   | 207                          | 5.28                                      | Measured (Ball Area)      | 108.8           | 33.2                  |
| 22 | B6400VB-350                                                                                                                                                                 | 4                   | 350                          | 7.19                                      | Measured (Ball Area)      | 79.9            | 24.4                  |
| 23 | B6600VB-507                                                                                                                                                                 | 6                   | 507                          | 15.56                                     | Measured (Ball Area)      | 38.9            | 11.3                  |
| 24 | B6800VB-865                                                                                                                                                                 | 8                   | 865                          | 25                                        | Measured (Ball Area)      | 23              | 7                     |
| 25 | B61000VB-1550                                                                                                                                                               | 10                  | 1550                         | 44.62                                     | Measured (Ball Area)      | 12.9            | 3.9                   |
| 26 | B61200VB-1905                                                                                                                                                               | 12                  | 1905                         | 64.55                                     | Measured (Ball Area)      | 8.9             | 2.7                   |

Figure 10 – The Velocity sheet, showing a completed outlet velocity check with a recommended valve and pipe-size warnings.

| Field / Dropdown          | Where it is                 | Choices                                                            |
|---------------------------|-----------------------------|--------------------------------------------------------------------|
| Valve Type                | Velocity sheet, cell C2     | Butterfly, Globe, HTCCV, V-Ball, VS, VSS                           |
| Steam Quality Override    | Velocity sheet, cell C3     | (blank = auto-classify), Dry Saturated, Wet Saturated, Superheated |
| Target Velocity %         | Velocity sheet, cell G4     | 50, 80, 120                                                        |
| Supply / Outlet Pipe Size | Velocity sheet, cells J2–J3 | 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12 (inches)    |

Enter the steam flow, select a valve type and pipe sizes, and click Calculate Outlet Velocity. The sheet reports velocity in ft/s and m/s and the resulting Mach number for both the selected valve and the inlet/outlet piping, and flags any pipe that's running faster than the practical velocity limit for a short run — exactly as shown in the worked example above, where both the inlet and outlet 2-inch Schedule 40 pipe were flagged as running above the general guideline, with a note that this can still be acceptable for a short pipe run per some engineering standards.

## 11. The Engineering Behind the Tool, in Plain Language

You don't need to understand the math to use the tool — but if you want to explain a result to a customer, or you're the kind of person who likes to know what's happening under the hood, here's the short version of what the tool is actually doing.

### 11.1 Why pressure drop (not just flow) sizes the valve

A valve's whole job is to create a controlled restriction. The bigger the pressure drop you allow across the valve (inlet pressure minus outlet pressure), the less restrictive (and therefore smaller) the valve needs to be to pass a given flow. The tool's core formula is a version of the standard valve-sizing equation:

$$C_v = Q \div (1.62 \times \text{inlet pressure, absolute}) - \text{the choked-flow form (see 11.2)}$$

In everyday terms: more flow needs a bigger valve; more inlet pressure lets you use a smaller one. Below the choke point (11.2) the pressure drop across the valve matters as well.

he

### 11.2 Choked flow — what it means and why the tool warns you about it

Steam (like any compressible gas) can only accelerate through a restriction so fast. Once the pressure drop across the valve gets large enough — specifically, once the outlet pressure drops below roughly 55–58% of the inlet absolute pressure (the exact number depends on the valve) — the steam velocity at the narrowest point (the vena contracta) hits the local speed of sound and cannot go any faster no matter how much you lower the downstream pressure further. This condition is called choked flow, and the pressure ratio where it starts is the critical pressure ratio.

#### Why this matters to you

Once flow is choked, dropping the downstream pressure further doesn't get you more flow through the same valve — so the tool calculates the valve size using the critical pressure drop, not your full entered pressure drop, whenever it detects choked conditions. If you see a message like “Critical pressure drop reached,” that's the tool telling you it switched to this more conservative calculation to keep the valve sizing accurate, and also flagging that the installation may be noisier or more erosive at that condition — worth a second look, especially on larger pressure drops.

### 11.3 Valve rangeability and why it matters for turndown

Rangeability is the ratio between the largest and smallest flow a valve can control accurately — a valve with 50:1 rangeability can control flow accurately anywhere from 100% down to 2% of its rated capacity, but below that it starts to lose precision. If your building's actual minimum load is a much smaller fraction of peak than the valve's rangeability allows for, that's when a split-range (1/3-2/3) pair, covered in Section 5, becomes worth considering.

### 11.4 Equal-percentage valve characteristic

Most of the valves in this product line have an “Equal %” flow characteristic, meaning that equal increments of valve stroke (position) produce equal percentage changes in flow, rather than equal absolute changes. This characteristic pairs well with modulating (throttling) control because it helps keep the loop's overall gain more consistent across the full range of operation — which is part of why “Modulating” control (Section 3.2) generally points toward this valve family.

### **11.5 Unit conversions, handled for you**

Whichever units you enter your flow, pressure, and results in, the tool converts everything to a common internal basis (lb/hr and PSIA) before running the sizing math, then converts the results back to your chosen display units. This is why you can mix, for example, a flow given in MBH with a pressure given in BAR without doing any manual conversion yourself.

## 12. Troubleshooting & Getting Help

### 12.1 “Enable Content” bar won't go away / buttons don't do anything

This almost always means macros are still disabled. Close the file, reopen it, and click “Enable Content” on the yellow security bar as soon as it appears (Section 2.1). If your organization's IT policy blocks macros entirely, the tool will not be able to calculate — ask your IT team to allow this specific file, or run it from a trusted location.

### 12.2 The valve I picked manually was rejected

The tool checks the pressure rating of any manually selected valve (Section 3.7.1) against your entered inlet pressure. If you see a warning message, the valve you picked isn't rated for the job's pressure — choose a different model, or double check that your inlet pressure was entered correctly and in the right unit (PSIG vs. BAR).

### 12.3 My Cv/Kv label on the split-range sheet looks wrong

See the callout in Section 3.6 — this is a known cosmetic display issue on the OneThirdTwoThird sheet's label only. The calculated numbers themselves are correct.

### 12.4 Something else went wrong / unexpected error

If the tool shows an unexpected error message, note what you were entering and what the message said, then use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1). The more specific you are, the faster we can track it down.

### 12.5 Who to contact

For questions this manual doesn't answer, use the Report a Bug or Give Feedback button on the Calculator sheet (Section 4.1), or reach out to your Belimo Regional Application Consultant directly.

## Appendix A — All Dropdowns, One Table

A single quick-reference table of every dropdown in the tool, where to find it, and every choice it offers.

| Field / Dropdown                  | Where it is           | Choices                                                                                       |
|-----------------------------------|-----------------------|-----------------------------------------------------------------------------------------------|
| Application                       | Calculator! D6        | Coil, Fin Tube, Heat Exchanger, Radiator                                                      |
| Control                           | Calculator! D8        | On/Off, Modulating                                                                            |
| Flow Unit                         | Calculator! D12       | BTU/HR, EDR, LB/HR, MBH, KG/h                                                                 |
| Inlet Pressure Unit               | Calculator! D13       | PSIG, BAR                                                                                     |
| Outlet Pressure Unit              | Calculator! D14       | PSIG, BAR                                                                                     |
| Location (Barometric Pressure)    | Calculator! D15       | Default, Manual / Other, all 50 U.S. states + Canadian provinces (74 total — see table below) |
| Coefficient Type                  | Calculator! D28       | Cv, Kv                                                                                        |
| Selected Valve (manual override)  | Calculator! J23       | -Select-, plus all 59 Belimo steam valve models (see the Valves sheet, Section 9)             |
| Target Velocity (ft/s)            | Calculator! C38       | 50, 80, 120                                                                                   |
| Velocity Check – Valve Type       | Calculator! H39       | Butterfly, Globe, HTCCV, V-Ball, VS, VSS                                                      |
| Velocity Check – Steam Type       | Calculator! H40       | Dry Saturated, Wet Saturated, Superheated                                                     |
| Velocity Check – Pipe/Valve Sizes | Calculator! H41–H43   | -Select-, 0.5–12 in. (14 sizes)                                                               |
| View                              | Calculator! C47       | Simple, Detailed                                                                              |
| Suggested Split Valves            | OneThirdTwoThird! C25 | -Select-, Ball, Globe                                                                         |
| Valve Type                        | Velocity! C2          | Butterfly, Globe, HTCCV, V-Ball, VS, VSS                                                      |
| Steam Quality Override            | Velocity! C3          | (blank = auto), Dry Saturated, Wet Saturated, Superheated                                     |
| Target Velocity %                 | Velocity! G4          | 50, 80, 120                                                                                   |
| Supply/Outlet Pipe Size           | Velocity! J2–J3       | 0.5–12 in. (14 sizes)                                                                         |

### Location / Barometric Pressure — full list

The Location dropdown (Calculator! D15) offers the following 74 entries:

Default • Manual / Other • Alabama • Alaska • Arizona • Arkansas • California • Colorado • Connecticut • Delaware • Florida • Georgia • Hawaii • Idaho • Illinois • Indiana • Iowa • Kansas •

Kentucky • Louisiana • Maine • Maryland • Massachusetts • Michigan • Minnesota • Mississippi • Missouri • Montana • Nebraska • Nevada • New Hampshire • New Jersey • New Mexico • New York • North Carolina • North Dakota • Ohio • Oklahoma • Oregon • Pennsylvania • Rhode Island • South Carolina • South Dakota • Tennessee • Texas • Utah • Vermont • Virginia • Washington • West Virginia • Wisconsin • Wyoming • ...and additional Canadian provinces / territories